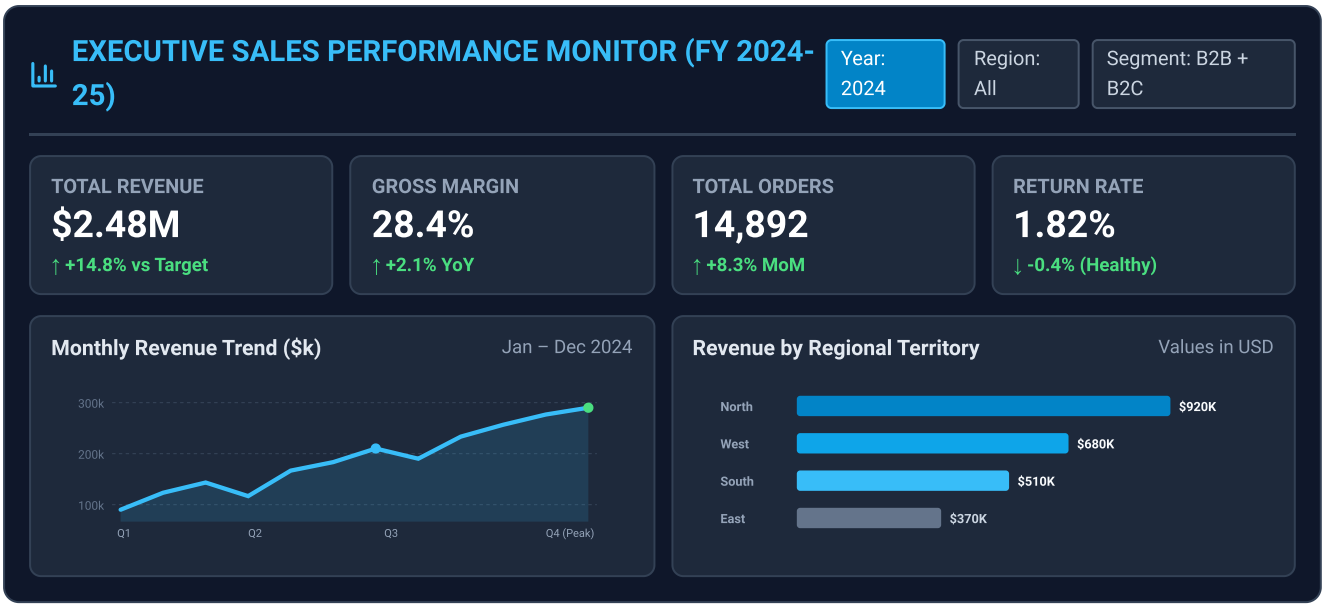
EXECUTIVE PRESENTATION & ACADEMIC PROJECT DELIVERABLE

Power BI Enterprise Sales & Operations Dashboard

Designed for Corporate Decision Making • 3-Member Collaborative Presentation Guide

Platform: Microsoft Power BI Desktop | **Team Size: 3 Members** | **Time Allocation: 8–10 Minutes Total** | **Audience: Academic Faculty & Evaluators**



Team Project Breakdown (3-Person Split)

Speaker 1: Project Scope

Introduces problem statement, business objective, data architecture, and top-level KPI metrics.

Speaker 2: Deep Dive

Walks through data charts, seasonal revenue patterns, and performs dynamic slicer filtering on screen.

Speaker 3: Strategic Value

Explains executive business recommendations, operational ROI, project conclusions, and leads viva defense.

TECHNICAL ARCHITECTURE & METHODOLOGY

How the Dashboard Was Built: Step-by-Step

Explain this section when the teacher asks: *"How did you create this dashboard?"* or *"Explain your ETL and data modeling pipeline."*

1. **Data Sourcing & Extraction (Get Data)**
   Imported multiple operational tables from Excel and CSV formats into Power BI Desktop, including `SalesOrders`, `CustomerMaster`, `ProductCatalog`, and `RegionMapping`.

2. **Data Cleaning & ETL in Power Query Editor**
   Handled missing values, converted currency formats to numeric values, removed duplicate order records, established standard date formatting, and created custom conditional columns for regional tiers.

3. **Data Modeling & Star Schema Design**
   Organized tables into a robust Star Schema. Placed `Fact_Sales` at the center and connected surrounding dimension tables (`Dim_Date`, `Dim_Product`, `Dim_Region`) using **1-to-Many (1:*) relationships** with single-direction cross filtering.

4. **Calculations & DAX Measures (Data Analysis Expressions)**
   Wrote dedicated business measures rather than relying on default sums to guarantee accurate aggregations across any applied filter context:

KPI Metric	DAX Formula Used	Business Purpose
Total Revenue	`Total Sales = SUM(Fact_Sales[SalesAmount])`	Aggregates gross billed value.
Gross Profit Margin	`Margin % = DIVIDE([Total Profit], [Total Sales], 0)`	Evaluates net unit profitability.
YoY Growth Rate	`YoY % = DIVIDE([Total Sales] - [PY Sales], [PY Sales])`	Measures annual expansion velocity.
Active Customers	`Active Clients = DISTINCTCOUNT(Fact_Sales[CustomerID])`	Prevents duplicate customer counts.

Examiner Defense Tip: If asked why you used `DIVIDE()` instead of a simple forward `/`, say: *"The DIVIDE function handles division-by-zero errors safely by returning zero or null instead of crashing the dashboard."*

DELIVERY SCRIPT • TEAM HANDOVER

Presentation Speaking Script: Part 1

Speak naturally with steady pacing. Point to the physical dashboard screen during key calls.

Speaker 1: Introduction, Problem Statement & KPIs

Duration: 2.5 Minutes

Formal Greeting & Team Welcome:
*"Respected faculty members and fellow classmates, a very good morning. Today, our team is presenting our Business Analytics project: **Interactive Enterprise Sales Intelligence Dashboard using Microsoft Power BI**. My name is [Speaker 1 Name], and presenting with me today are my colleagues, [Speaker 2 Name] and [Speaker 3 Name]."*

The Business Problem Statement:
"In fast-paced corporate environments, managing operations through static Excel spreadsheets creates severe data silos, slow reporting turnaround, and human calculation errors. Top executives waste hours waiting for weekly consolidation reports. Our objective was to develop a single, real-time interactive intelligence dashboard that delivers instant operational visibility."

Walking Through Top KPI Cards:
*"Looking at the top tier of our dashboard, we have engineered four critical executive indicators: **Total Revenue ($2.48M)** reflecting a healthy 14.8% surplus over planned quota; our **Gross Margin (28.4%)**; over **14,800 fulfilled orders**; and a tightly controlled **1.82% return rate**."*

Handover Callout to Speaker 2: "These top cards summarize our overall business health. I will now hand over to [Speaker 2 Name], who will explain our trend analysis, regional distributions, and demonstrate dynamic slicers."

Speaker 2: Visual Analysis, Seasonality & Dynamic Slicing

Duration: 3.0 Minutes

Takeover & Visual Breakdown:
*"Thank you, [Speaker 1]. Good morning everyone, my name is [Speaker 2 Name]. Moving to our graphical visual tier: our **Monthly Sales Line Chart** reveals clear quarter-over-quarter expansion, reaching peak sales velocity during Q4 holiday demand."*

Regional Comparison & Live Demo:
*"On the right, our **Regional Bar Chart** illustrates that the **North Territory leads with $920K**, generating nearly 37% of aggregate turnover. Conversely, the East region represents an underutilized market at $370K. What makes Power BI truly powerful is dynamic filtering: if I select 'Region: North' or toggle our '2024' slicer, the underlying DAX calculations cross-filter every metric on the canvas instantaneously without requiring a manual refresh."*

Handover Callout to Speaker 3: "Having reviewed how the visuals operate interactively, [Speaker 3 Name] will now interpret the executive findings, strategic recommendations, and conclude our presentation."

STRATEGIC INSIGHTS & EXAM VIVA DEFENSE

Presentation Speaking Script: Part 2 & Viva Q&A

Final closing script for Speaker 3 followed by the exam defense cheat sheet for the whole group.

Speaker 3: Business Insights, Recommendations & Conclusion

Duration: 2.5 Minutes

Opening & Core Management Findings:

"Thank you, [Speaker 2]. Hello everyone, I am [Speaker 3 Name]. In business analytics, visual charts are only valuable if they lead to profitable management decisions. From our analysis, we recommend three strategic actions:

- ***Resource Reallocation:*** *Double promotional investment in the high-margin North and West zones to capitalize on existing sales momentum.*
- ***Supply Chain Correction:*** *Re-evaluate product mix and distribution in the East zone to reverse low performance.*
- ***Operational Efficiency:*** *Automating this workflow eliminates approximately 15 to 20 manual hours of monthly data consolidation."*

Formal Project Conclusion & Q&A Transition:

"In conclusion, Power BI converts raw, fragmented transaction files into clear commercial strategy. This project helped our team bridge academic theory with real enterprise analytics. We thank our respected faculty for their mentorship. We are now open for questions and feedback. Thank you!"

Viva Voce / Teacher Defense Cheat Sheet

Examiner Question	Primary Responder	Recommended Clear Answer
"Why choose Power BI instead of Microsoft Excel?"	Speaker 1 / 2	*"Excel slows down significantly with large datasets and lacks automated drill-down capabilities. Power BI utilizes a high-performance in-memory tabular engine (VertiPaq) and offers interactive multi-chart cross-filtering."*
"What is the difference between a Calculated Column and a Measure?"	Speaker 2 / 1	*"A Calculated Column consumes RAM by storing a row-by-row value inside the table. A DAX Measure is calculated on-the-fly only when referenced in a visual, preserving system memory."*
"What concrete business benefit does this deliver to a CEO?"	Speaker 3	*"It eliminates delayed retrospective reporting. An executive can identify revenue leakages, high-margin categories, and regional deficits in seconds rather than waiting for month-end reports."*

Exam Scoring Tip: Stand tall, maintain eye contact with the examiners, and pass questions naturally using phrasing like: *"My teammate [Name] handled that specific data model layer and will answer that."*